WEEK 10 - WEEK 12

Objectives:

1. Finalize data analysis and interpretation.
2. Critical evaluation of the data analysis.
3. Prepare presentation slides for project seminar presentation. Discuss with your supervisor.

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WEEK 10 - WEEK 12

Summary of weekly activities

Finalization of Data Analysis and Interpretation

During this reporting period, the data analysis phase was finalized by executing a rigorous hyperparameter tuning sweep on the 1,000 simulation acoustic dataset generated via MATLAB Simscape. The analytical focus was directed toward optimizing the Long Short-Term Memory (LSTM) neural network to process raw, 1,500-step time-series sequences of pipeline pressure data. The final interpretation of the training progress confirmed that a specific architectural configuration, utilizing the Adam optimizer, 125 hidden units, a mini-batch size of 16, a 0.001 learning rate, and a cap of 50 epochs successfully allowed the model to converge. This configuration established a strict mathematical balance, enabling the network to identify the Negative Pressure Wave (NPW) without overfitting to the background SCADA noise, ultimately yielding a robust final validation accuracy of 70.33%.

Critical Evaluation of the Data Analysis

A critical evaluation of the data processing methodologies was conducted, uncovering significant engineering insights regarding signal integrity and model complexity. Initial testing explored the use of moving average smoothing filters to eliminate sensor static; however, live diagnostic testing revealed that this filtering acted destructively by mathematically flattening the sharp inflection point of the NPW. Consequently, the methodology was revised to process strictly raw, unfiltered data. By applying a high Dropout rate of 0.5, the burden of noise filtration was successfully shifted to the LSTM’s native "Forget Gate," which preserved the physical integrity of the wave. Furthermore, a comparative evaluation against a Bidirectional LSTM (BiLSTM) demonstrated that the standard, unidirectional LSTM offered superior industrial stability, as it was not mathematically misled by the chaotic, post-leak wave reflections that disrupted the BiLSTM's backward pass.

Preparation of Seminar Presentation Slides and Supervisor Consultation

In preparation for the final project seminar defense, all analytical findings and system developments including the interactive digital twin dashboard were synthesized into a comprehensive presentation deck. The presentation was structured to strictly address the FYP Part 2 marking rubric, clearly outlining the problem statement, simulation methodology, performance benchmarking, and real-time dashboard capabilities. Following a direct review and consultation session with the project supervisor, critical refinements were integrated into the deck. These improvements included adding direct citations and pros/cons to the literature review, reordering the presentation flow for better logical sequencing, explicitly detailing the

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tuned hyperparameters in the architecture section, and incorporating the training and validation loss graphs to visually defend the model's generalization and lack of overfitting.

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